Communities and Mobile Services in African Society: The Agricultural Cooperative Sector Opportunity

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Abstract

This paper presents an overview of the landscape of agriculture cooperatives in Sub-Saharan Africa and opportunities for mobile services. This study analyses the key activities and existing mobile and ICT services for cooperatives distributed over two main service segments, information and financial services.

Our study highlights that cooperatives are an essential pillar of the agriculture domain. While today very few mobile and ICT services are targeting these organizations, the study shows that there is not only a great potential but also a strong demand for such services, in particular those that could be used by the vast majority of farmers, i.e. that are based on voice channel and basic phones. Cooperatives and their members represent also a great environment for adoption of mobile money services. There is a huge potential to propose bundled financial services through e.g. a partnership with Micro-Finance Institutions (MFIs). However, this would require mobile money platform to be accessible to farmers and cooperatives in a form that they can easily and universally use.

1. Introduction

This paper presents the results and findings of a research study conducted from March 2013 till November 2013 by the authors in Sub-Saharan Africa. The major objective of this study was to explore the potential of mobile services for cooperatives of the agriculture sector in Sub-Saharan Africa. The aim was to investigate how cooperatives work, what the profiles of the different actors (internal and external actors from the cooperative perspective) are, and how the existing m-Agri (mobile-based agriculture services) portfolio of services currently deployed in different countries fits with these entities.

2. Methodology

The study was developed in three stages. The first stage was focused on a desk research on mAgri services, and cooperatives separately to understand how these entities work, what their importance for the sector is, what the services they provide are, and what ICT services are currently addressing their needs or the needs of their members. In terms of needs, the study explored both information requirements and financial requirements. The second stage consisted of a field study and interviews with a wide range of actors in a representative set of countries from East and West Africa. Interviewed actors included cooperative managers, ICT service providers, buyers and traders, regional actors and media actors (community radio stations). Finally, the third stage aimed at consolidating the desk research and the field study, and identifying key outputs and results presented in this paper.

3. Cooperatives overview

Over several decades Cooperatives have played an important role in society on several fronts. A significant piece of this is related to unlocking the economic potential, particularly for the low-income earning smallholder producer population. Owing to their structure as being transparent, democratic and self-governed, cooperatives offer the institutional framework through which local communities can earn their livelihoods (OFEIL, 2005)[1]

More than 40% of all households in Africa are members of a cooperative society.[2] African cooperatives, like any other have created a significant economic value by the sheer number of jobs created by the sector. Cooperatives establish links between the informal and the formal sectors and generate economies of scale and scope through horizontal and vertical integration.

[1] OFEIL. (2005). Participative schemes and management structures of Ghanaian Cooperatives. *Journal of Co-operative Studies, Vol. 38:3, No. 115*, 14-26.
[2] Source: ILO: http://www.microfinancegateway.org/gm/document-1.9.28915/22563_cooperatives.pdf

3.1 Classification of Cooperatives Found Across Africa[3]

1. **Worker-owned**: Cooperatives provide their members with jobs
2. **Member-owned**: People group together to conduct group activities together. Are predominant in the agricultural sector, can stabilize existing self-employment in rural areas
3. **Financial cooperatives**: Mobilize savings among the poorest and thus accumulate capital for productive investment
4. **Social cooperatives**: Provide self-employed workers with a minimum of social security while creating jobs in the social service sector

ILO in its study (2008) identifies two main types of cooperatives found in the ten countries it studied. These include:

1. **Agricultural cooperatives:** Sometimes called multipurpose cooperatives, rural cooperatives and marketing cooperatives, among others.
2. ***Saving / rotating and credit cooperatives (SACCOs / ROSCAs):*** Formal and informal saving groups.

3.2 Activities

The main activities of a coop can be divided under two major segments:-

Internal: This relates to all the activities a coop has to perform internally and in close cooperation with its members.

- Support of Production:
  a. Purchasing of inputs (seeds, fertilizers, agriculture implements etc.)
  b. Training
  c. Sharing tools
  d. Dissemination of information (weather forecast, etc.)
- Trade support
  a. Transporting
  b. Marketing
  c. Group selling
  d. Dissemination of information (market prices, etc.)
- Financing: A combination of the above also involve financial inputs like farm credit, seed purchase credit etc.

External: This relates to all the activities a coop has to perform with the external actors who are not the active / registered members of the coops. These can be

- Traders looking to do business with the coop
- Agents representing a business or several businesses

[3] Derived from ILO: http://www.microfinancegateway.org/gm/document-1.9.28915/22563_cooperatives.pdf

- Input sellers, e.g. shopkeepers, agents of agri companies
- Service providers, e.g. financing companies / agents

Within the two major segments above there can be several tasks associated with running the coop. A typical coop task matrix is depicted below:

Running the Coop (All Internal)

- Encouraging people to sign up as members and collecting fees and subscriptions ;
- Providing members with information and services
- Collecting feedback from members

Supporting Production

- Provide members with technical advice and information (e.g. daily/weekly rainfall totals, agri techniques, pest and soil management, post-harvest steps etc.)
- Provide reminders and alerts and warnings (e.g. locusts, severe storms, draught etc.) (again internal and external)
- Advertise prices then collect and aggregate orders from members for inputs (seeds, fertiliser, pesticides) (internal and external)
- Place orders, receive inputs from suppliers; deliver inputs to members (external)
- Collect payments for inputs (internal)
- Collect information from members on progress through agricultural season; pests and diseases seen; quantities harvested and for sale (internal)
- Reply to members questions on relevant topics or other concerns (internal)

Supporting Trade (external)

- Staying informed about market trends (demand and prices) both locally and globally
- Sharing market information with members and identifying new opportunities to trade
- Negotiating deals and securing finance
- Informing producers of orders secured (quantities, qualities and price)
- Collecting information from producer about quantities available
- Organising quality control and aggregation and packaging of production
- Organising transport and delivery of orders to clients
- Making payments to individual producers

Within the ambit of internal and external activities two major segments emerge

1. **Information based services**: as identified in the distribution above notwithstanding the distinction between internal and external there is a specific segment of activities that are purely information based. These include e.g. market

price collection and dissemination, access to MIS[4] data, weather services, advice and information related to farm inputs like seeds, fertilizer etc.

2. **Financial services**: this segment caters to accounting, farm finance, input finance, insurance, transport services etc. where either money inflow is required or is disbursed to the coop members.

Note on a key actor

Women are a cornerstone of African economic development, providing an estimated 70 percent of agricultural labour and producing between 78 and 90 percent of all food – and they do so on subsistence and small land holdings, owning very little land or other productive resources. Hence it is important to keep this important segment in mind when designing ICT services for the coop sector.

4. Challenges faced by cooperatives

4.1 The urban rural divide

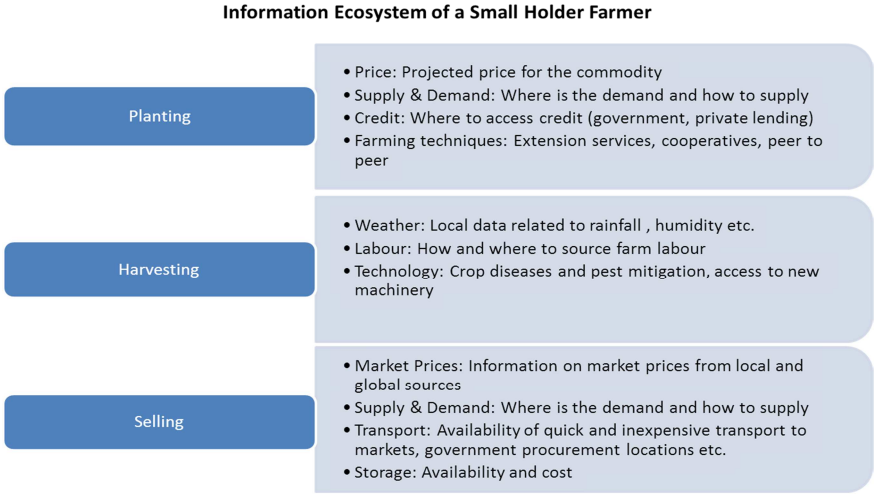
There are very specific challenges related to the urban and rural divide when it comes to agri-coops. Usually the urban counterparts are much more equipped with resources-linked to both internal and external activities like access to markets, agri experts and private companies supplying farm inputs and buyers etc. The rural coops struggle to match resources and also lack the exposure to buyers as their urban counterparts do.

4.2 Internal Challenges

Information related

The following depicts a traditional information ecosystem for a smallholder farmer who can be part of a small or large coop.

[4] Market Information System



Derived from Web for Regreening in Africa Research Initiative http://www.w4ra.org

Farmer coops have traditionally relied on a range of information sources; however the **information thus provided may be outdated, biased or inadequate**. The challenges identified (see table below) with traditional sources of information can potentially have negative consequences for farmer coops and their chief source of income.

Source	Advantages	Disadvantages
Knowledge from peers & village community	▪ Native knowledge hence trusted ▪ Best practices and evidence of success available locally and passed on from generations	▪ Limited knowledge of latest farming technology, outcomes and crop related information ▪ Risk of inadequate and outdated information being passed on across the farmer value chain
Farmer societies & cooperatives	▪ Trusted as one of their own ▪ Provide a one stop shop for information, tools and technology for a wide range of services	▪ Limited knowledge of latest farming technology, outcomes and crop related information ▪ Limited to their own sector of work, e.g. seeds or fertilizer only.
Local shopkeeper	▪ Trusted and the most visited source for farmers ▪ Local knowledge and availability of products and solutions at farmers doorstep ▪ Provides informal access to credit	▪ Bias of information towards sale of products for maximizing profit ▪ Inadequate knowledge of best agriculture information and products
Private companies	▪ Providers of solutions/ products and related education, not Information alone	▪ For profit motivation resulting in product / technology bias ▪ More focus on retail than quality of information
NGOs	▪ Free and locally available services ▪ Usually by a member of the community hence trusted	▪ Limited knowledge of latest farming technology, outcomes and crop related information
Government extension office	▪ Free training and education services ▪ Government agricultural extension services are trusted ▪ Assumed to be always available in time of need	▪ Lack of availability at the time of need ▪ Due to a significant gap of resources the government extension worker may not always be equipped with locally relevant information ▪ Gender issues

Derived from Web for Regreening in Africa Research Initiative – http://www.w4ra.org

Finance related

For smallholder farmers access to capital is always a challenge. Coops have an essential function to keep its members financially comfortable by either providing pre-financing options or post-harvest assured sale. However in predominantly rural areas such linkages, especially to the Micro Finance Institutions is a challenge, and usually difficult for coops

to manage remotely. It is completely different set of opportunities and access to finance related information and services in the urban context. The larger urban coops do have decent and robust technology systems for financial accounting, pre-finance etc.

4.3 External Challenges

Information Related

Lack of access to buyers both globally and locally and lack of reliable pricing data in the marketplaces are common information related external challenges. Due to its silo approach existing systems do not really connect the buyers and coops effectively. At most they have created efficiency at the middle level where the processing of intermediaries has become more efficient

Finance Related

Access to capital and markets to sell is the primary challenge facing agri-coops. In marketing terms low commodity prices, operational difficulties, the agricultural economy, and competition are the major challenges facing this sector.

5. ICT's existing channels, tools, challenges and needs

It is evident that ICT's and especially the mobile phone have huge potential in offering tools and services to this sector. The table below summarizes what has already been deployed and tested:-

Channel	Tool / Tech	Applications (few popular ones)	Target (end user)	Challenges
Mobile Phone	SMS	Market price, weather, insurance, marketing messages etc. **Ranking:** Very popular as a deployment channel	Literate coops members and smallholder farmer	Non-literate members cannot access enriched information like , weather information, buyer contacts etc.
	Call-Centre	Helplines, MIS, Marketing etc. **Ranking:** Getting popular as a deployment channel	Literate & non-literate coops members and smallholder farmer	Sustainability challenges for the operator
	IVR	Helplines, Market price, weather, insurance, marketing messages etc. **Ranking:** Still in maturity phase as a deployment channel	Literate & non-literate coops members and smallholder farmer	IVR can cope with literacy issues but as people have no prior history of using such interfaces, design and usability are key challenges
	Mobile Web	Audio, visual and textual content along with helplines **Ranking:** Still in early phase as a deployment channel	Literate coops members and smallholder farmer	Either non-existent or very expensive for rural consumers & non-literate cannot access
TV	Video	Training modules, market information, weather information, advertisements **Ranking:** Very popular as a deployment channel backed mostly by governments or donors	Literate & non-literate coops members and smallholder farmer	Content needs to be interactive enough and in local language for mass consumption, not accessible in rural areas Electricity and low penetration of such services add to the challenges related to accessibility

Radio	Audio	Communiques, advertisements, announcements, Q&A, market price **Ranking:** Mostly in early stages as a deployment channel	Literate & non-literate coops members and smallholder farmer	Cost of broadcast, availability of infrastructure for the radio stations Lack of ability of people to pull content as they may miss key broadcast owing to their daily work schedule
Web	Text/A/V	Audio, visual and textual content along with refined agri content **Ranking:** Getting popular as a deployment channel in urban context	Literate coops members and smallholder farmer	Either non-existent or very expensive for rural consumers

The following aspects represent the key takeaways in addition to tools related challenges

- **Usability:** Majority of the initiatives targeted at the agri sector rely on the use of SMS as a channel for information dissemination. While technologically this suffices for the market information to flow from an aggregator to the end consumer, it is severely **challenged on the aspects of literacy and the languages supported**. Poor uptake of SMS based services indicates preference for low-cost, easy to use, locally relevant information source. While the advent of mobile phone has laid down the framework of a potential revolution in the communications space people in rural areas still have significant challenges accessing services that require textual literacy.
- **Infrastructure**: depending on urban/rural segment access to information channels like TV, Radio, and Internet may be a challenge. Availability of electricity may also be a significant challenge.
- **Service**: Lack of services aimed at the coops e.g. stock and financial management in rural areas where there is no connectivity or infrastructure
- **Applications**: lack of applications linked to the locally available infrastructure.

5.1 Mobile Money

Through the Mobile Phone channel, Mobile Money (MM) services are increasingly playing a part in the socio-economic landscape of least developed and developing countries. These services enable people to transact either in cash-in cash-out scenarios or a simple call time transfer over a mobile network operator. In unstructured market conditions where a Point of Sale (PoS) may not necessarily have an electronic payment solution at hand Mobile Money can play a significant role in coop activities. This is equally applicable to the internal coop member transactions also.

However there are several challenges associated with the use of Mobile Money. Our research highlights three major challenges can impact a more global adoption of MM in coops:

1. **Awareness:** Concerning awareness, it is essential to inform potential users about the major advantages of MM. This element is essential in societies where cash is the essence of financial transactions.
2. **Training:** The use of technologies such as SMS and USSD is not widespread in rural areas. The lack of knowledge of how to use mobile money, as well as the lack of training is one of the major barriers that prevent greater adoption.
3. **Accessibility:** This is a key issue in rural areas. Given the usual greater illiteracy rate (e.g. in Mali >80% in rural area), and the low usage of technologies such as SMS or USSD, the current MM clients are not accessible for a large part of the rural communities.

However just like the ubiquitous usefulness of radio based services in such rural areas MM holds great promise provided the above challenges are taken care of. In some countries like Afghanistan mobile operators[5] are already making novel use of mobile money. E.g. disbursing salaries in areas that are hard to get to and where corruption in payments is a chronic problem.

6. Gap Analysis

We can summarize the gaps under technology / information gaps and access gaps (can be market or physical access)

6.1 Technology related

- *Service Structure issues:* Low literacy level in rural areas, the inability for SMS to support non-Latin characters, and the user interface that prevents people who are literate to use a 9 key keyboard to enter text are major inhibitors of information access for the underprivileged population. Members of rural coops are usually the smallholder farmers who usually have a low literacy level, no prior exposure to technology, and speak languages that are have not been used before in the ICT services targeted at them.
- *Unmet potential of Web based services:* While there are several ICT based initiatives to serve the smallholder farmer segment, there are unique challenges that are now widely acknowledged. For example a service like MFarm in Kenya has a web based interface that serves literate and connected traders, but that same interface cannot help illiterate and not so well connected smallholders.

6.2 Access Related

- *Information needs and gender asymmetry:* In the case of sub-Saharan African, only 69% of female farmers receive visits from agriculture extension agents. The extension

[5] http://world.time.com/2013/03/02/how-afghanistan-is-on-the-leading-edge-of-a-tech-revolution/

agents who actually conduct farm visits are usually male members (e.g. 97% of the farm visits are conducted by male extension agents).

- *Information has empowered intermediaries:* Due to socio-economic factors the intermediaries have had the first mover advantage. A remarkable majority of market information systems are led by this new class of info-mediary hindering a direct coop to buyer interaction.
- *Infrastructure and access issues:* For the rural coops access to regional markets and large/global traders is imperative. The mobile phone is the most promising tool that is available at a relatively low cost. However existing systems do not really connect the buyers and sellers effectively. At most they have created efficiency at the middle level where the processing of intermediaries has become more efficient as mentioned above.
- *Market Asymmetry:* As the agri- coop does not have access to market prices (daily or seasonal) its ability to (a) communicate; (b) negotiate directly with the buyer is a challenge.

7. Findings

The study, particularly through the interviews, has highlighted a very interesting point. While agriculture is a current major focus of the mobile and ICT industry, and while the so-called m-agriculture is attracting more and more attentions from development actors, all the initiatives we have identified are targeting farmers directly, and we couldn't find any examples of services trying to empower these existing organizations.

The technology component that can play a very significant part here is the new paradigm of introducing Mobile Money services both at the cooperative and member level as well as the business end of cooperatives. One of the key limitations for mobile money usage is related to users' challenges. The study shows that there are three major issues related to accessibility, awareness, and training that are heavily impacting a greater adoption of mobile money service.

Right set of incentives must exist for coop members. A link with MFI's or lending institutions with seamless interoperability can boost the trust and ease of use of such services. The study has also highlighted the importance of financial transactions between members, as well as the critical role of cooperative for accessing loans by members. All these functionalities (financial transactions, targeted credit, warehouse receipt, etc.) could be easily handled through Mobile Money.

In terms of business partnerships for both buying and selling, the visibility of cooperatives as well as other stakeholders is a major issue. Cooperatives usually rely a lot on local links for input buying and product selling. They have hardly access to regional, national or international markets for their operations, creating major differences in prices between regions or places. Some cooperatives are able to develop their global links but

this requires investments. Here there is very little use of ICT to identify opportunities and develop partnership.

Service Based Opportunities (o be provided by mobile operators / other players) to cooperatives

- **Reliable and recent market prices** to empower farmers in their negotiation with middlemen. Lots of cooperatives in our study are developing themselves an ad-hoc network of contact from which they can gather the information in an informal way and then share with their members. In this area, trust in the collected information is essential.
- **Weather forecast adapted to the crops**. The usual public weather information is not adapted to farmers need, and acquiring relevant information for harvesting or planting is critical, but also difficult to achieve.
- **Access to new markets for both buying & selling** (business matching). Access to new market opportunities is a key challenge. Knowing where is the demand and where the offer is critical. Lots of farmers are limiting their productions due to lack of selling opportunities. The role of the cooperative in that domain is essential, even if, as mentioned earlier, access to market is also a challenge for the organization. At the same time, for some crops, the demand is unmet in other places in the region or the country.
- **Exchange of information between the cooperatives and their members**. Cooperatives and their members have lots of information to share from alerts (new group buying or group selling opportunities), to e.g. meetings (e.g. training). The information flow both direction from members to cooperatives and vice-versa. While person-to-person mobile phone connections have eased a lot these exchanges, the use of more advanced ICT services, particularly for asynchronous communications, can largely improve current situations.

For all these cooperative-internal services, it is critical to mention that all interviews have highlighted the fact that SMS and other text channels are not accessible to most of farmers, who are highly illiterate, and have only the ability to dial a phone number. In such context, the only accessible technology is likely to be voice-based services that need to be carefully designed to be accessible to people without any experience with ICT.

Concerning services for cooperatives to connect with their external partners (suppliers, traders, buyers), we have identified a set of functionalities that could leverage access to new opportunities:

- **Product advertisement:** Cooperatives should have an easy way to promote the products and services they are providing or have available at a given time. The key

element here is to ensure that the cooperatives are in charge of updating the information to be sure it is always up to date.

- **Buying requests:** In the same way, cooperatives are buying inputs, and placing requests for specific demand could be a way to have access to better opportunity.
- **Automatic Notifications:** for the two above mentioned functionalities, it is essential to have a notification system, probably based on SMS. While cooperatives, buyers, traders, sellers have access to Web platform, they would be interested to get notified when specific offer or demand are reaching the platform.
- **Advertisement of cooperatives:** Apart from just selling and buying products, cooperatives are interested to promote themselves, increase their visibility and highlight their specificities (fair trade, organic product, women-produced products, etc.). It would be therefore very interesting to have a kind of yellow pages for cooperatives and other actors of the domain to increase mutual knowledge and visibility.

While mobile phones have been leveraging communication between members, the knowledge sharing is heavily linked to personal networks. There is a clear demand for easy-to-use tools that would ease and support the links between members. However, such tools must take into account the capacities of farmers to use technology.

8. Conclusion

In conclusion this study has highlighted the importance of cooperatives as federating organizations for farmers. Our study highlights that there is a strong demand for ICT tools to empower the cooperatives / NGO's. However, it is likely that for the moment IVR-based tools are the most capable to address the requirements of the farmers. It is also important to note that cooperatives and their members would be a great environment for adoption of mobile money services. There is a huge potential to propose bundled financial services through a partnership with MFIs. However, this would be possible and should be tested only when mobile money will be accessible to famers and cooperatives.

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